Magnetic Levitation at a Nominal 10 mm Distance

Abridged Calculation: Position PI Driving PWM Directly

ENGR 155 calculation notes

Result. Reducing the distance improves the available lifting force. For the proposal's illustrative coil, a 10 mm *center distance* requires about 1.178 A and 78.5% ideal duty at 12 V. However, assuming an inductance of 10 mH and negligible passive mechanical damping, the position PI loop remains unstable for every choice of PI gains in the continuous-time model derived below. The inductance and damping are assumptions, not measurements.

1 Geometry, architecture, and assumptions

The suspended magnet, carrier, and payload have total mass $m = 0.010\,\mathrm{kg}$. The desired *surface gap* is $d_0 = 0.010\,\mathrm{m}$, measured from the lower coil face to the upper magnet face. The thin-coil/dipole model instead uses the distance z from the coil plane to the magnet center:

$$z = d + \Delta_g, \qquad z_0 = d_0 + \Delta_g. \tag{1}$$

For a coil represented by its axial midplane and a magnet represented by its geometric center, Δ_g is approximately half the coil thickness plus half the magnet height. Its value must be determined from the actual geometry.

Distance convention for the numbers below. To reproduce the current 10 mm calculation, the numerical illustration sets $z_0 = 10\,\mathrm{mm}$. This equals a 10 mm surface gap only in the zero-offset, thin-coil/point-magnet idealization. For a real 10 mm surface gap, substitute the actual $z_0 = 10\,\mathrm{mm} + \Delta_g$ into the same equations.

The controller is *position PI only*: measured gap error produces PWM duty directly. There is no current reference or current-feedback controller. Current can still be measured for field compensation, telemetry, and protection. The FPGA executes the position PI and PWM; the MCU handles ADC acquisition, DMA, the gap estimate, and the web interface through the Wi-Fi module.

The numerical sizing assumptions from the proposal are

Symbol	Meaning	Value used
m	Total suspended mass	$0.010\,\mathrm{kg}$
g	Gravitational acceleration	$9.81\,\mathrm{m/s^2}$
N	Coil turns	500
a	Mean coil radius	$0.010\,\mathrm{m}$
μ	Magnet dipole moment	$0.050\,\mathrm{A\,m^2}$
R	Coil resistance	$8.0\,\Omega$
V_s	Supply voltage	$12.0\,\mathrm{V}$
L	Differential coil inductance	$0.010\,\mathrm{H}$, assumed
c	Passive mechanical damping	$0\,\mathrm{N\,s/m}$, approximation

The model assumes an aligned permanent dipole, a thin air-core coil, small motions near equilibrium, and an averaged PWM voltage $V_s u$ in continuous conduction. It neglects driver and diode drops, thermal changes, magnetic saturation, hysteresis, and unmodeled structural motion. A final core-equipped coil must be characterized locally rather than assigned these air-core slopes.

2 Holding force, current, and power

The standard axial coil field and aligned-dipole approximation give

$$B(z) = \frac{\mu_0 N I a^2}{2(a^2+z^2)^{3/2}}, \tag{2}$$

$$F(z,I) \simeq \mu \left|\frac{\mathrm{d}B}{\mathrm{d}z}\right| = K(z) I, \tag{3}$$

$$K(z) = \frac{3\mu_0 N \mu a^2 z}{2(a^2+z^2)^{5/2}}, \qquad \mu_0 = 4\pi \times 10^{-7}\,\mathrm{H/m}. \tag{4}$$

The field expression follows the current-loop model in OpenStax [1]; the force relation is the dipole approximation used in the proposal. At $z_0 = 0.010\,\mathrm{m}$,

$$K(z_0) = 0.0833041\,\mathrm{N/A}, \tag{5}$$

$$F_0 = mg = 0.0981\,\mathrm{N}, \tag{6}$$

$$i_0 = \frac{mg}{K(z_0)} = 1.17761\,\mathrm{A}, \tag{7}$$

$$u_0 \simeq \frac{R i_0}{V_s} = 0.78508, \tag{8}$$

$$P_{\mathrm{coil}} = i_0^2 R \simeq 11.09\,\mathrm{W}. \tag{9}$$

The ideal full-on current is $V_s/R = 1.50\,\mathrm{A}$, so this nominal point is statically feasible in the ideal model. It leaves about 21.5 percentage points of upward duty headroom; it does not establish thermal acceptability, dynamic force headroom, or closed-loop stability.

3 Small-signal force gains and mechanics

Define $x = d - d_0$ and $j = i - i_0$. Positive x is downward motion, away from the coil. A first-order Taylor expansion of upward magnetic force is

$$F \simeq mg - k_x x + k_f j, \qquad k_f = \left.\frac{\partial F}{\partial i}\right|_0, \qquad k_x = -\left.\frac{\partial F}{\partial d}\right|_0. \tag{10}$$

Because $z = d + \Delta_g$ with fixed geometry, $\mathrm{d}z/\mathrm{d}d = 1$. For the illustrative force law,

$$k_f = K(z_0), \tag{11}$$

$$\frac{\mathrm{d}K}{\mathrm{d}z} = K(z)\left(\frac{1}{z} - \frac{5z}{a^2+z^2}\right), \tag{12}$$

$$k_x = mg\left(\frac{5z_0}{a^2+z_0^2} - \frac{1}{z_0}\right). \tag{13}$$

Substitution gives

$$\boxed{k_f = 0.0833041\,\mathrm{N/A}, \qquad k_x = 14.715\,\mathrm{N/m}.} \tag{14}$$

Newton's law, with downward displacement positive, is

$$\boxed{m\ddot{x} + c\dot{x} - k_x x + k_f j = 0.} \tag{15}$$

If current were held fixed and $c = 0$, the mechanical poles would be $s = \pm\sqrt{k_x/m} = \pm 38.36\,\mathrm{s}^{-1}$. The positive pole represents a growing displacement. This is a fixed-current mechanical comparison, not the complete voltage-driven plant.

4 Position PI and the uncontrolled coil current

Let $u \in [0, 1]$ be duty fraction. For a constant target define

$$\boxed{u = u_0 + K_p x + K_i \eta, \qquad \dot{\eta} = x.} \tag{16}$$

The positive sign increases attraction when the object is too far from the coil. K_p has units m^{-1}, K_i has units $\mathrm{m}^{-1}\mathrm{s}^{-1}$, and η has units $\mathrm{m\,s}$. For a changing requested gap $r(t)$, the controller error is $e = d - r$.

Coil current is a plant state, even though it has no controller. Including motion-induced voltage, the local electrical equation is

$$\boxed{L\dot{j} + Rj - k_f\dot{x} = V_s\delta u, \qquad \delta u = u - u_0.} \tag{17}$$

The coupling term follows from reciprocal magnetic coenergy: the downward magnetic force is $-F$, so $\partial\lambda/\partial d = \partial(-F)/\partial i = -k_f$, where λ is flux linkage. Consequently $\delta v = Rj + L\dot{j} - k_f\dot{x}$. This is the energy-based electromechanical modeling method in Woodson and Melcher [2]. The mechanical damping c excludes this explicitly modeled coil coupling. If motion-induced voltage is negligible, omit the $-k_f\dot{x}$ term; then the $+k_f^2$ term below also disappears.

5 Laplace transform and stability calculation

With zero initial conditions, the transformed equations are

$$(ms^2 + cs - k_x)X + k_f J = 0, \tag{18}$$

$$(Ls + R)J - k_f sX = V_s\Delta U, \tag{19}$$

$$\Delta U = \left(K_p + \frac{K_i}{s}\right)X. \tag{20}$$

Eliminating J and ΔU gives

$$s\big[(ms^2 + cs - k_x)(Ls + R) + k_f^2 s\big] + k_f V_s(K_p s + K_i) = 0. \tag{21}$$

Thus the characteristic polynomial is

$$\boxed{\begin{aligned} p(s) = mLs^4 + (mR + cL)s^3 + (cR - k_x L + k_f^2)s^2 \\ + (k_f V_s K_p - k_x R)s + k_f V_s K_i. \end{aligned}} \tag{22}$$

The four states are position, velocity, current, and accumulated position error. Taking a transform changes the representation, not the physical stability. Laplace and feedback foundations are in Oppenheim and Willsky [4].

For $p(s) = A_4 s^4 + A_3 s^3 + A_2 s^2 + A_1 s + A_0$, strict stability requires

$$A_4, A_3, A_2, A_1, A_0 > 0, \tag{23}$$

$$A_3 A_2 - A_4 A_1 > 0, \tag{24}$$

$$A_3 A_2 A_1 - A_4 A_1^2 - A_3^2 A_0 > 0. \tag{25}$$

These are the quartic Routh–Hurwitz conditions [3]. Two necessary conditions are

$$K_p > \frac{k_x R}{k_f V_s}, \qquad cR + k_f^2 > k_x L. \tag{26}$$

The second condition is independent of the PI gains.

For $L = 10\,\mathrm{mH}$ and $c = 0$, the numerical polynomial is

$$\boxed{\begin{aligned} p(s) &= 0.0001 s^4 + 0.08 s^3 - 0.140210 s^2 \\ &\quad + (0.999649 K_p - 117.72) s + 0.999649 K_i. \end{aligned}} \tag{27}$$

Its s^2 coefficient is negative for every K_p, K_i. Therefore no position PI tuning can make this particular continuous-time model strictly stable. The necessary proportional-gain threshold, $K_p > 117.76\,\mathrm{m^{-1}}$, cannot repair the negative coefficient.

For illustration only, choose $K_p = 150\,\mathrm{m^{-1}}$ and $K_i = 100\,\mathrm{m^{-1}s^{-1}}$. Then

$$p(s) = 0.0001 s^4 + 0.08 s^3 - 0.140210 s^2 + 32.2273 s + 99.9649, \tag{28}$$

whose roots are

$$\boxed{s = -802.247, \quad -2.9963, \quad 2.6214 \pm j\,20.2237\ \mathrm{s^{-1}}.} \tag{29}$$

The oscillatory mode grows as $e^{2.6214t}$, with an amplitude e-folding time of about 0.381 s. These are *unstable example gains*, not recommended firmware settings.

The necessary electrical/damping condition can also be written

$$c = 0: \quad L < \frac{k_f^2}{k_x} = 0.4716\,\mathrm{mH}, \tag{30}$$

$$L = 10\,\mathrm{mH}: \quad c > \frac{k_x L - k_f^2}{R} = 0.01753\,\mathrm{N\,s/m}. \tag{31}$$

Passing either applicable threshold is necessary but not sufficient: all Routh inequalities must still pass. This does not prove that every physical PI-controlled levitator is unstable; it identifies what this assumed plant cannot achieve.

6 Digital implementation and the z-transform

For an illustrative 1 kHz position update, $T = 0.001\,\mathrm{s}$, choose the explicit timing convention

$$\delta u[k] = K_p e[k] + K_i \eta[k], \tag{32}$$

$$\eta[k+1] = \eta[k] + T e[k]. \tag{33}$$

This gives $C(z) = K_p + K_i T/(z-1)$. Using the updated accumulator in the same sample instead would give a different discrete controller.

To check the held actuator correctly, define $q = [x, \dot{x}, j]^{\mathsf{T}}$:

$$\dot{q} = Aq + B\delta u, \quad A = \begin{bmatrix} 0 & 1 & 0 \\ k_x/m & -c/m & -k_f/m \\ 0 & k_f/L & -R/L \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ V_s/L \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}. \tag{34}$$

The exact zero-order-held plant has

$$A_d = e^{AT}, \qquad B_d = \int_0^T e^{A\tau} B \, \mathrm{d}\tau, \tag{35}$$

and the sampled closed-loop state matrix, with zero additional computation delay, is

$$M = \begin{bmatrix} A_d + B_d K_p C & B_d K_i \\ TC & 1 \end{bmatrix}. \tag{36}$$

The implemented loop is stable only if every eigenvalue of M is inside the unit circle. Merely substituting $z = e^{sT}$ for the continuous closed-loop poles does not model a separately sampled controller and held actuator. Additional sensor filtering, communication, and computation delay must be included before selecting firmware gains.

For the same illustrative plant and the unstable example gains above, the eigenvalues of this sampled matrix are

$$z = 0.447767, \quad 0.997019, \quad 1.002880 \pm j\, 0.020241. \tag{37}$$

The complex pair has magnitude approximately 1.00308, exceeding one. Thus the explicitly defined 1 kHz sampled implementation is also unstable.

7 Measurements and the position-tolerance requirement

For the final coil at the actual 10 mm surface gap, measure the holding current and hot resistance, differential inductance, and local force slopes:

$$k_f \simeq \frac{F(d_0, i_0 + \Delta i) - F(d_0, i_0 - \Delta i)}{2\Delta i}, \tag{38}$$

$$k_x \simeq -\frac{F(d_0 + \Delta d, i_0) - F(d_0 - \Delta d, i_0)}{2\Delta d}. \tag{39}$$

Estimate passive damping without counting the modeled coil-induced damping twice. Recalculate the continuous and sampled poles with the measured values. Duty limits and integrator anti-windup are necessary practical details, but they do not turn an unstable linear PI design into a stable small-signal loop.

Specify a true-position tolerance and a recovery time separately; for example, $|d - r| \le 0.5\,\mathrm{mm}$ after a defined disturbance and settling interval. That example is not a demonstrated performance claim. A stable, unsaturated PI loop can reject constant disturbances in its measured position, while sensor bias, noise, quantization, and saturation still limit the true position. The proposal's current demonstration band is $\pm 5\,\mathrm{mm}$. No settling time or position band can be certified from the unstable example above.

References

[1] OpenStax, *University Physics, Volume 2*, Sec. 12.4, "Magnetic Field of a Current Loop." https://openstax.org/books/university-physics-volume-2/pages/12-4-magnetic-field-of-a-current-loop

[2] H. H. Woodson and J. R. Melcher, *Electromechanical Dynamics*, Part I, Chs. 3 and 5. MIT OpenCourseWare. https://ocw.mit.edu/courses/res-6-003-electromechanical-dynamics-spring-2009/pages/part-i/

[3] University of Illinois, ECE 486 Handbook, Lecture 7, "Routh–Hurwitz Stability Criterion." https://courses.physics.illinois.edu/ece486/sp2026/documentation/handbook/lec07.html

[4] A. V. Oppenheim, A. S. Willsky, with S. H. Nawab, *Signals and Systems*, 2nd ed., Prentice Hall, 1997, Chs. 9 (Laplace), 10 (z-transform), and 11 (feedback). MIT reading guide: https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/pages/readings/